

Foundations in Data Science Final Project

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Introduction

This is a final project submission for the course Foundations in Data Science. Within this project, subject RNA-seq count and meta data from patients enrolled in The Cancer Genome Atlas (TCGA) Breast Invasive Carcinoma (TCGA-BRCA) cohort was examined. The ten genes specifically chosen throughout this project were done so due to their known influence on breast cancer risk. Information provided by the National Breast Cancer Foundation was utilized to choose eight of these genes, with the other two found by a general search. The primary gene analyzed in the figures and tables is TACSTD2. This is the gene that provides instructions for making the Trop-2 protein, which is highly expressed in many solid tumors and is associated with cancers, including breast cancer.

Methods

The analysis performed in this project was completed using R version 4.5.1.

The packages utilized in R include:

ggplot2 4.0.0 Used to generate all of the plots within this project

harrypotter 2.1.1 Used in the violin plot for the color scheme

ComplexHeatmap 2.24.1 Used to generate the heat map

circlize 0.4.16 Used to create certain elements in the heat map

dplyr 1.1.4 Used for formatting data to plot, particularly in the heat map

Results

The most notable trend derived from multiple plots is how, particularly in the Violin Plot and the Density Plot, the furthest and most severe phase represented in the chosen covariate, Stage IV, contains the highest count or expression of the TACSTD2 gene. This is an outcome that aligns with what is known about this gene and its relationship to breast cancer. Although there are significant outliers, the majority of the genes analyzed stayed within a lower count, or expression, range. Seen most clearly in the heat map, only a few of the genes chosen are expressed in a higher range, which was interesting as all of the genes were chosen due to their known relationship with breast cancer.

Figures and Tables

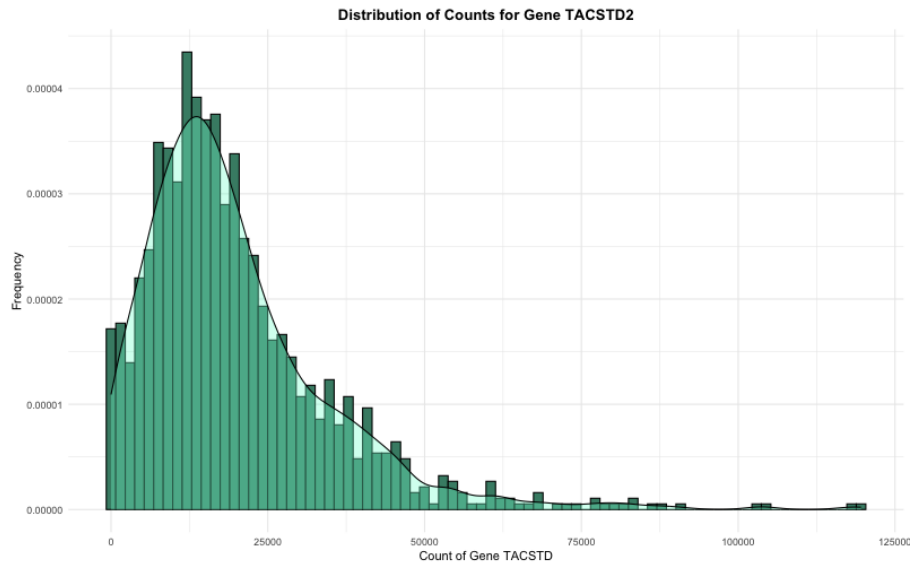


Figure 1: The histogram is displaying the distribution of gene expression for the TACSTD2 gene, including a distribution curve. There are multiple outliers shown in the right tail, but the data appears to be relatively consistently distributed near a count of 2000, which correlates with the mean as seen in the table below.

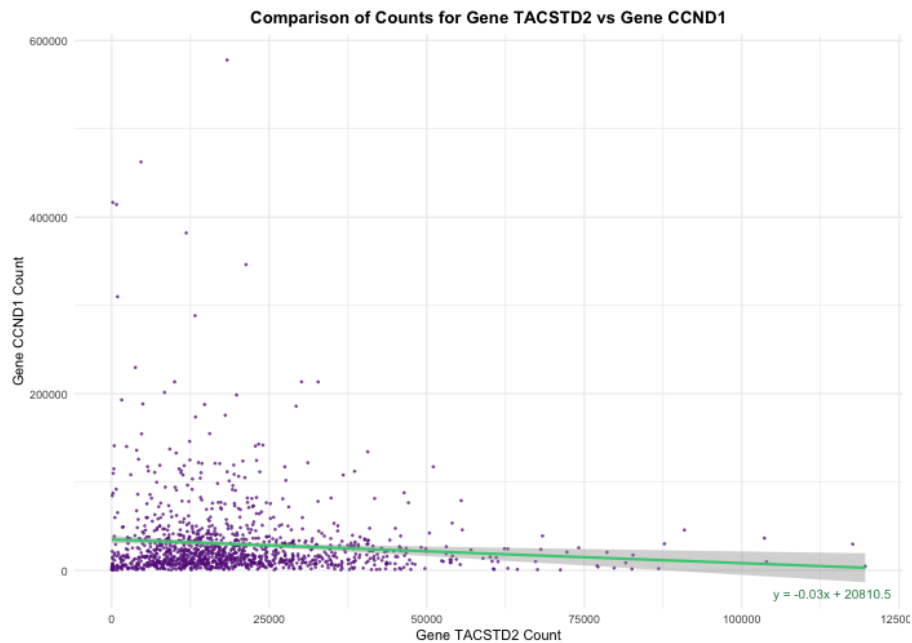


Figure 2: The scatter plot displays a comparison of gene expression for the TACSTD2 and CCND1 genes. There is a very slight negative correlation, but, overall, not a significant, distinguishable relationship.

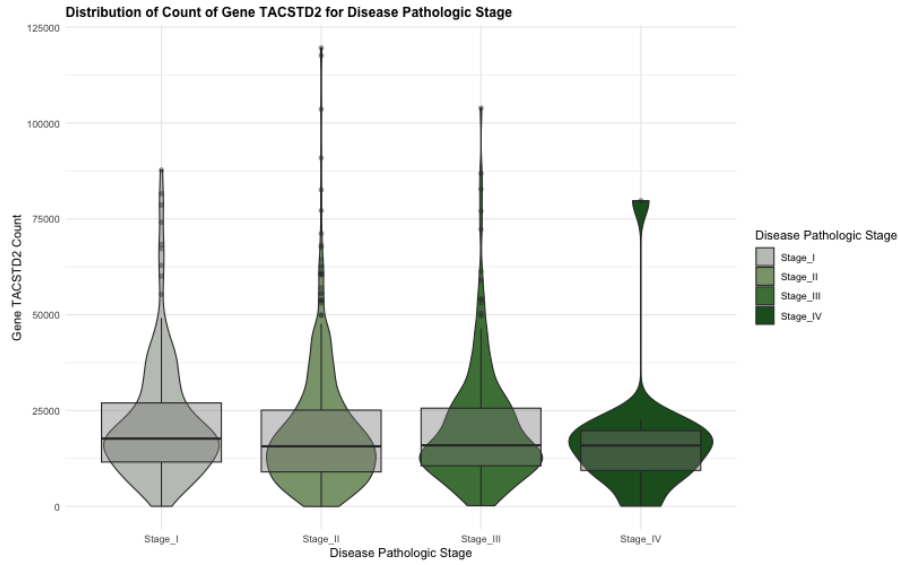


Figure 3: The violin plot, including inserted box plots, displays the distribution of gene expression for the TACSTD2 gene by pathologic stage. The furthest stage represented here, Stage IV, does have the highest distribution of increased expression, which is an expected outcome.

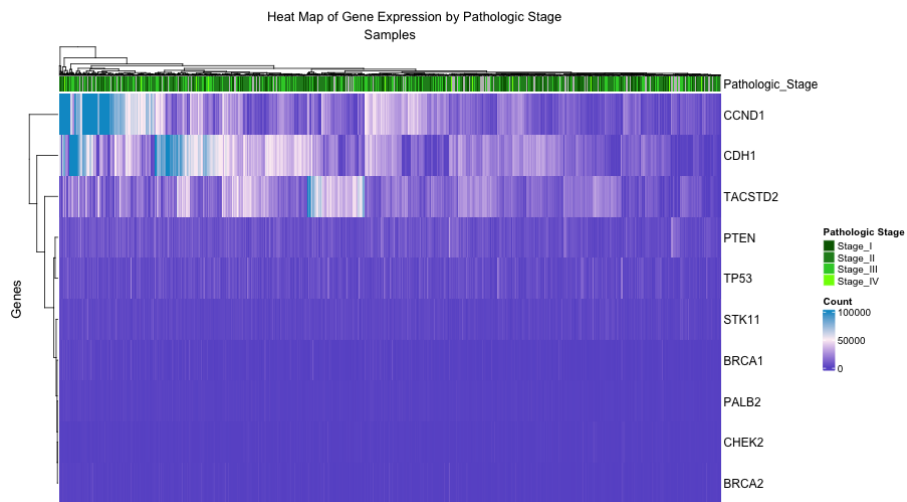


Figure 4: The heat map displays counts of the listed ten genes and their pathologic stage for each sample. With the large number of subjects, it is difficult to draw significant findings from this map, but most notably, the first three genes listed display the highest expression counts.

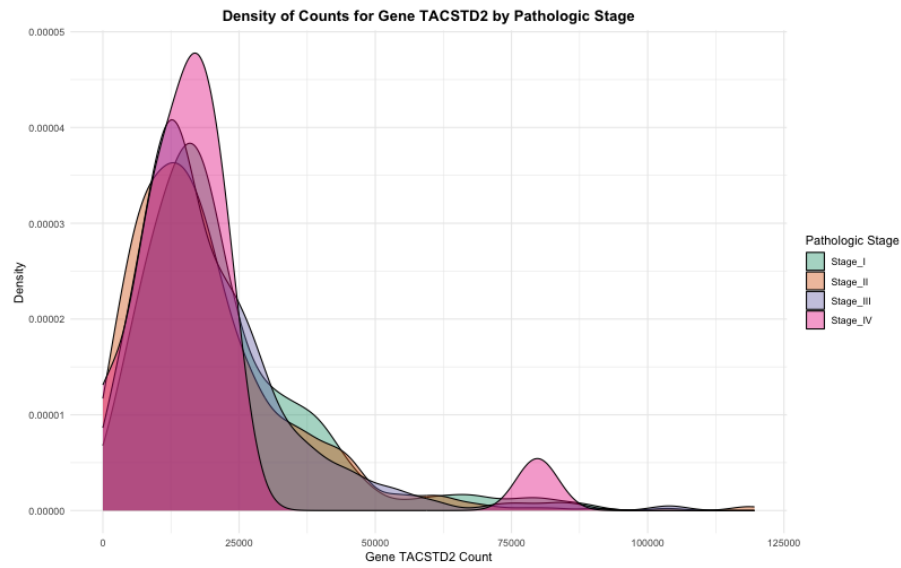


Figure 5: This density plot, added for the final submission, is displaying the expression of the TACSTD2 gene by pathologic stage. Phase IV appears to have the greatest distribution of higher count or expression.

Summary Statistic	TACSTD2	CCND1
Minimum	1	430
Median	16523	16473
Mean	19878	29649
Maximum	119596	577835
Standard Deviation	15161.47	44263

Table 1: Summary Statistics for Specified Genes generated using base R functions `summary()` and `sd()`

Acknowledgments

The results shown here are in whole or part based upon data generated by the TCGA Research Network:
<http://cancergenome.nih.gov/>.

References

- [1] Lingle, W., Erickson, B. J., Zuley, M. L., Jarosz, R., Bonaccio, E., Filippini, J., Net, J. M., Levi, L., Morris, E. A., Figler, G. G., Elnajjar, P., Kirk, S., Lee, Y., Giger, M., and Gruszauskas, N. (2016). The Cancer Genome Atlas Breast Invasive Carcinoma Collection (TCGA-BRCA) (Version 3) [Data set]. The Cancer Imaging Archive.
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- [2] Other Breast Cancer Genes,
<https://www.nationalbreastcancer.org/other-breast-cancer-genes/>
- [3] Tables,
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